

Assignment 3 - Programming Error Analysis Sheet

Question No.	Error	Line No.	Error Type	Error Correction
1	Missing semicolon after int x = 10	8	Syntax Error	int x = 10;
1	Assignment operator = instead of comparison == in if	13	Logical Error	if(x == 10)
1	String assigned to int variable	18	Type Error	int marks = 90;
1	Undefined variable z	20	Semantic Error	int z = 0; printf("%d\n", z);
1	Lvalue required - 100 = x is invalid	22	Syntax Error	x = 100;
1	Array index out of bounds (arr[10] for size 5)	26	Runtime Error	arr[4] = 50;
1	Incompatible pointer type - float* to int*	28	Type Error	int *ptr = &x;
1	Wrong format specifier %f for int	30	Type Error	printf("%d\n", x);
1	Division by zero	35	Runtime Error	Check if b != 0 before division
1	Multi-character constant 'AB'	37	Type Error	char ch = 'A';
1	Same message in if and else (logical error)	39	Logical Error	if(x > y) printf("X is greater"); else printf("Y is greater");
1	sqrt() return value unused (warning)	44	Logical Error	int result = sqrt(25);
1	Undefined function show()	46	Semantic Error	Define show() or remove call
1	Semicolon after while makes empty loop	48	Logical Error	while(x < 15) { x++; }

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1	continue outside loop	53	Syntax Error	Remove or place inside loop
1	break outside loop/switch	55	Syntax Error	Remove or place inside loop/switch
1	Function calculate() called with 1 argument instead of 2	57	Semantic Error	<code>int result = calculate(10, 20);</code>
2	Missing semicolon after <code>int num2 = 20</code>	9	Syntax Error	<code>int num2 = 20;</code>
2	Missing closing parenthesis in if	21	Syntax Error	<code>if(choice == 1)</code>
2	Extra closing parenthesis in else if	33	Syntax Error	<code>else if(choice == 3)</code>
2	Missing closing parenthesis in for loop	50	Syntax Error	<code>for(int i=0; i<5; i++)</code>
2	Missing closing parenthesis in while	57	Syntax Error	<code>while(choice > 0)</code>
2	Missing semicolon after do-while	66	Syntax Error	<code>while(choice > 5);</code>
2	Missing colon after case 2	74	Syntax Error	<code>case 2:</code>
2	Missing closing quote in char literal	101	Syntax Error	<code>char grade = 'A';</code>
2	Missing closing bracket in array declaration	105	Syntax Error	<code>int arr[5];</code>
2	Missing closing brace for block	126	Syntax Error	Add <code>}</code> before <code>printf("Still Running")</code>
2	Missing semicolon after printf	133	Syntax Error	<code>printf("End of Program\n");</code>
3	String assigned to int variable	8	Type Error	<code>int age = 20;</code>
3	String assigned to	14	Type Error	<code>char grade =</code>

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	char (should be single quotes)			'A';
3	Wrong format specifier %d for float	16	Type Error	printf("%f\n", salary);
3	Wrong format specifier %f for int	18	Type Error	printf("%d\n", marks);
3	Undefined variable score	22	Semantic Error	int score = 0;
3	Variable x redeclared	25	Semantic Error	x = 20;
3	Lvalue required - 100 = x	27	Syntax Error	x = 100;
3	Wrong format specifier %s for int	31	Type Error	printf("%d\n", num);
3	Wrong format specifier %d for float	35	Type Error	printf("%f\n", pi);
3	Floating point array index	39	Type Error	arr[2] = 100;
3	Float assigned to char (truncation)	41	Type Error	char ch = 'A';
3	Function add() called with 1 argument instead of 2	45	Semantic Error	result = add(10, 20);
3	Function average() called with 2 arguments instead of 3	51	Semantic Error	avg = average(10, 20, 30);
3	String assigned to int	55	Type Error	int temperature = 35;
3	String assigned to float	59	Type Error	float discount = 10.5;
3	Float subtracted from int (implicit conversion)	61	Type Error	amount = amount - (int)discount;
3	Array name cannot be assigned	65	Syntax Error	strcpy(name, "Robil");

Question No.	Error	Line No.	Error Type	Error Correction
	directly			
3	Undefined variable totalMarks	71	Semantic Error	int totalMarks = 0;
3	Wrong format specifier %c for int	75	Type Error	printf("%d\n", number);
3	Wrong format specifier %d for float	79	Type Error	printf("%f\n", rate);
3	Function add() called with 3 arguments instead of 2	86	Semantic Error	sum = add(a, b);
3	Float result stored in int (truncation)	92	Type Error	float area; area = 3.14 * radius * radius;
3	Assignment = instead of comparison == in if	100	Logical Error	if(studentMarks == 100)
3	String assigned to int	105	Type Error	int size = 10;
3	Wrong format specifier %d for float	111	Type Error	printf("%f\n", percentage);
3	String used as array index	115	Type Error	arr2[5] = 100;
3	String literal in case label	125	Type Error	case 2:
3	Undefined function multiply()	132	Semantic Error	Define multiply() or remove call
3	Wrong format specifier %d for float	141	Type Error	printf("%f\n", areaRect);
3	Float returned but stored in int	151	Type Error	float avgMarks; avgMarks = average(...);
3	Undefined	157	Semantic Error	int value = 0;

Question No.	Error	Line No.	Error Type	Error Correction
	variable value			
3	String assigned to int	159	Type Error	char employeeId[] = "EMP101";
4	Function declared but not defined: calculateSalary	3	Semantic Error	Define calculateSalary()
4	Function declared but not defined: printStudentDetails	6	Semantic Error	Define printStudentDetails()
4	Function declared but not defined: showResult	9	Semantic Error	Define showResult()
4	Function declared but not defined: calculateAverage	10	Semantic Error	Define calculateAverage()
4	Extern variable totalStudents declared but not defined (extern needs definition elsewhere)	12	Semantic Error	Remove extern or define in another file
4	Extern variable companyProfit declared but not defined	13	Semantic Error	Remove extern or define in another file
4	Undefined function generateReport()	47	Semantic Error	Define generateReport()
4	Undefined function displayEmployee()	49	Semantic Error	Define displayEmployee()
4	Undefined function processData()	51	Semantic Error	Define processData()
4	Undefined function saveRecord()	53	Semantic Error	Define saveRecord()
4	Undefined function loadRecord()	55	Semantic Error	Define loadRecord()
4	Undefined function	57	Semantic Error	Define printInvoice

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	printInvoice()			()
4	Undefined function calculateTax()	59	Semantic Error	Define calculateTax()
4	Undefined function validateUser()	61	Semantic Error	Define validateUser()
4	Undefined function sendNotification()	63	Semantic Error	Define sendNotification()
4	Undefined function generateCertificate()	65	Semantic Error	Define generateCertificate()
5	Division by zero	10	Runtime Error	Check if b != 0 before division
5	Dereferencing NULL pointer	13	Runtime Error	if(ptr1 != NULL) *ptr1 = 50;
5	Array index out of bounds	18	Runtime Error	arr[4] = 100;
5	Dereferencing uninitialized pointer	21	Runtime Error	int *ptr2 = malloc(sizeof(int)); *ptr2 = 200;
5	Buffer overflow - strcpy into small array	25	Runtime Error	char name[20]; strcpy(name, "Programming");
5	Dereferencing uninitialized pointer	29	Runtime Error	int *ptr3 = malloc(sizeof(int)); printf("%d\n", *ptr3);
5	File not found - fopen returns NULL	33	Runtime Error	if(fp != NULL) { fscanf(...); fclose(fp); }
5	Buffer overflow in malloc'd memory	38	Runtime Error	char *ptr4 = malloc(20); strcpy(ptr4, "Computer

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				Science");
5	Use after free	43	Runtime Error	Don't access memory after free
5	Double free	48	Runtime Error	Free only once
5	Modulo by zero	53	Runtime Error	Check if y != 0 before x % y
5	Buffer overflow - strcat into small array	57	Runtime Error	char str1[30]; strcat(str1, "WorldProgramming");
5	Array out of bounds in malloc'd memory	62	Runtime Error	numbers = malloc(10 * sizeof(int));
5	2D array index out of bounds	68	Runtime Error	matrix[2][2] = 100;
5	Dereferencing NULL pointer	71	Runtime Error	if(ptr6 != NULL) printf("%d\n", ptr6[0]);
5	Using gets() - unsafe function	75	Runtime Error	Use fgets(password, sizeof(password), stdin);
5	Use after free	80	Runtime Error	Don't access ptr7 after free
5	Using uninitialized variable age	84	Runtime Error	int age = 0;
5	Negative array index	87	Runtime Error	if(index >= 0) arr[index] = 50;
5	Dereferencing uninitialized pointer	90	Runtime Error	Initialize ptr8 before dereferencing
5	Dereferencing	94	Runtime Error	if(text !=

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	NULL string pointer			NULL) printf("%s\n", text);
5	Potential memory allocation failure	99	Runtime Error	if(hugeArray != NULL) hugeArray[0] = 1;
5	Infinite loop - pointer arithmetic on NULL	104	Runtime Error	Remove infinite loop
6	Wrong logic - odd printed as even	13	Logical Error	if(num % 2 == 0) printf("Number is Even\n"); else printf("Number is Odd\n");
6	Wrong grade logic - 90+ should be A not B	23	Logical Error	if(marks >= 90) printf("Grade A\n"); else if(marks >= 80) printf("Grade B\n");
6	Incomplete leap year logic	33	Logical Error	if((year % 4 == 0 && year % 100 != 0) (year % 400 == 0))
6	Wrong logic - age > 18 should be eligible	43	Logical Error	if(age >= 18) printf("Eligible for Voting\n"); else printf("Not Eligible for Voting\n");
6	Wrong logic - smaller number logic reversed	53	Logical Error	if(a < b) printf("Smaller Number = %d\n", a); else printf("Smal

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				ler Number = %d\n", b);
6	Wrong logic for largest - using < instead of >	63	Logical Error	if(y > largest) largest = y; if(z > largest) largest = z;
6	Integer division instead of float division	73	Logical Error	average = (float)total Marks / subjects;
6	Wrong formula - area uses 2 (circumference formula)	83	Logical Error	area = 3.14 * radius * radius;
6	Wrong bonus calculation - 0.02 should be 0.20 and vice versa	93	Logical Error	if(salary > 30000) bonus = salary * 0.20; else bonus = salary * 0.02;
6	Wrong month names - case 1 is January not February	103	Logical Error	case 1: printf("Janu ary\n"); case 2: printf("Febr uary\n"); .. .
6	Wrong prime logic - isPrime set to 1 when divisible	115	Logical Error	isPrime = 0; break;
6	Wrong discount logic - adding instead of subtracting	126	Logical Error	finalAmount = purchaseAmou nt - discount;
6	Wrong sum logic - sum overwritten instead of accumulated	136	Logical Error	sum += i;
6	Wrong temperature logic - < 40 is hot not cold	146	Logical Error	if(temperatu re > 35) printf("Hot weather\n");

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				else printf("Cold Weather\n");
6	Wrong grade logic - A should be excellent not average	156	Logical Error	if(grade == 'A') printf("Excellent Performance\n"); else printf("Average Performance\n");
6	Wrong area formula - using + instead of *	166	Logical Error	rectangleArea = length * width;
6	Wrong pass/fail logic - >= 60 should be pass	176	Logical Error	if(percentage >= 60) printf("Pass\n"); else printf("Fail\n");
7	Array index out of bounds - i<=5 for size 5	10	Runtime Error	for(int i=0; i<5; i++)
7	Integer division for average - should be float	20	Logical Error	average = (float)total / 5;
7	Array index out of bounds - i starts at 1, misses 0	27	Runtime Error	for(int i=0; i<10; i++) numbers[i] = i * 10;
7	Buffer overflow - strcpy into small array	37	Runtime Error	char name[20]; strcpy(name, "ComputerScience");
7	Array name cannot be assigned directly	42	Syntax Error	strcpy(city, "Delhi");
7	String comparison using == instead of strcmp	49	Logical Error	if(strcmp(str1, str2) == 0)
7	Using gets() - unsafe function	56	Runtime Error	Use fgets(password, sizeof(password),

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				ord), stdin);
7	Buffer overflow - strcat may exceed array size	64	Runtime Error	char firstName[40]; strcat(firstName, lastName);
7	String not null- terminated - grade array too small	71	Runtime Error	char grade[3] = "A+";
7	Negative array index	76	Runtime Error	arr[0]
7	Buffer overflow - email array too small	87	Runtime Error	char email[30] = "student@gma il.com";
7	Array index out of bounds - scores[3] for size 3	96	Runtime Error	int scores[4];
7	String comparison using == instead of strcmp	107	Logical Error	if(strcmp(us ername, "admin") == 0)
7	Buffer overflow - strcpy into small array	114	Runtime Error	char mobile[15]; strcpy(mobil e, "98765432101 2");
7	Array index out of bounds - i<=5 for size 5	121	Runtime Error	for(int i=0; i<5; i++)
7	Array index out of bounds - course[20] for size 20	128	Runtime Error	course[0] to course[19] only
7	String not null- terminated - college array	135	Runtime Error	char college[20]; college[0]=' A';... college[5]=' \0';
7	Buffer overflow - section array too small	149	Runtime Error	char section[5] = "IT-A";

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8	Function add() called with 1 argument instead of 2	14	Semantic Error	<code>result = add(num1, num2);</code>
8	Function average() called with 2 arguments instead of 3	20	Semantic Error	<code>avg = average(70, 80, 90);</code>
8	Factorial of 0 returns 0 instead of 1	26	Logical Error	<code>if(n == 0) return 1;</code>
8	Swap by value - doesn't actually swap x and y in main	35	Logical Error	Use pointers: <code>void swap(int *a, int *b)</code>
8	Maximum returns wrong value (returns smaller)	45	Logical Error	<code>if(a > b) return a; else return b;</code>
8	Fibonacci recursion wrong - fibonacci(n) instead of fibonacci(n-1)	55	Logical Error	<code>return fibonacci(n-1) + fibonacci(n-2);</code>
8	Undefined function calculateGrade()	62	Semantic Error	Define calculateGrade()
8	Undefined function circleArea()	67	Semantic Error	Define circleArea()
8	Undefined function sumArray()	72	Semantic Error	Define sumArray()
8	Undefined function displayResult()	77	Semantic Error	Define displayResult()
8	Wrong percentage formula - integer division first	102	Logical Error	<code>return (marks * 100) / total;</code>
8	Function returns float but no return value	108	Semantic Error	<code>return si;</code>
8	Power function	121	Logical Error	<code>for(int i=0;</code>

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	loop starts at 1 instead of 0			<code>i<exp; i++)</code>
8	reverseNumber doesn't return the reversed number	130	Semantic Error	<code>return rev;</code>
8	isPrime returns 1 for non-prime, 0 for prime (inverted)	145	Logical Error	<code>return 0; // for non-prime, return 1 for prime</code>
8	printArray uses <code><=size</code> causing out of bounds	160	Runtime Error	<code>for(int i=0; i<size; i++)</code>
8	Infinite recursion - recursiveDemo(n+1) never stops	165	Runtime Error	Add base case
8	Rectangle area returns sum instead of product	175	Logical Error	<code>return length * width;</code>
8	cube function returns <i>nn instead of nn*n</i>	180	Logical Error	<code>return n * n * n;</code>
9	Dereferencing uninitialized pointer	10	Runtime Error	<code>int *ptr1 = malloc(sizeof(int)); *ptr1 = 0;</code>
9	Dereferencing NULL pointer	14	Runtime Error	<code>if(ptr2 != NULL) *ptr2 = 100;</code>
9	Pointer arithmetic out of bounds	22	Runtime Error	Don't increment ptr3 beyond array
9	Array index out of bounds via pointer	27	Runtime Error	<code>*(ptr4 + 4)</code>
9	Use after free	37	Runtime Error	Don't access *ptr5 after free
9	Array out of bounds in malloc'd memory	46	Runtime Error	<code>ptr7 = malloc(10 * sizeof(int));</code>
9	Array out of bounds - ptr8[3] for malloc(3)	55	Runtime Error	<code>ptr8 = malloc(4 * sizeof(int))</code>

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				;
9	Double free	62	Runtime Error	Free only once
9	Dereferencing uninitialized malloc'd memory	67	Runtime Error	*ptr10 = 0; before printf
9	Array out of bounds - numbers[5] for malloc(5)	73	Runtime Error	for(int i=0; i<5; i++)
9	Swap function swaps pointers, not values	82	Logical Error	Use int temp = *a; *a = *b; *b = temp;
9	Double pointer used incorrectly - pptr should be int**	89	Type Error	int **pptr = &value_ptr; int *value_ptr = &value;
9	Buffer overflow - strcpy into small malloc	95	Runtime Error	name = malloc(20);
9	Pointer arithmetic out of bounds after increment	102	Runtime Error	Don't access *ptr11 after ptr11++
9	Dereferencing NULL pointer	107	Runtime Error	if(ptr12 != NULL && *ptr12 > 0)
9	Array out of bounds - ptr13[100] for malloc(100)	116	Runtime Error	ptr13[99] = 500;
9	Use after free	122	Runtime Error	Don't access *ptr14 after free
9	2D array accessed via pointer with wrong offset	129	Runtime Error	Use matrix[row][col] syntax
9	Memory leak - ptr15 malloc'd twice without freeing first	135	Runtime Error	free(ptr15); before second malloc or use realloc
10	Missing semicolon after int	8	Syntax Error	int choice = 1;

Question No.	Error	Line No.	Error Type	Error Correction
	choice = 1			
10	String assigned to int	10	Type Error	int marks = 90;
10	Function add() called with 1 argument instead of 2	16	Semantic Error	result = add(a, b);
10	Assignment = instead of comparison == in if	20	Logical Error	if(a == b)
10	Undefined variable score	23	Semantic Error	int score = 0;
10	Array index out of bounds	28	Runtime Error	arr[4] = 100;
10	Negative array index	30	Runtime Error	if(index >= 0) arr[index] = 50;
10	Buffer overflow - strcpy into small array	34	Runtime Error	char name[25]; strcpy(name, "Programming Language");
10	Array name cannot be assigned directly	39	Syntax Error	strcpy(city, "Delhi");
10	String comparison using == instead of strcmp	44	Logical Error	if(strcmp(user, "admin") == 0)
10	Division by zero	50	Runtime Error	if(y != 0) printf("%d\n", x/y);
10	Dereferencing NULL pointer	54	Runtime Error	if(ptr1 != NULL) *ptr1 = 500;
10	Dereferencing uninitialized pointer	58	Runtime Error	int *ptr2 = malloc(sizeof(int));
10	Use after free	64	Runtime Error	Don't access *ptr3 after free
10	Double free	68	Runtime Error	Free only once

Question No.	Error	Line No.	Error Type	Error Correction
10	Array out of bounds in malloc'd memory	72	Runtime Error	<code>ptr4 = malloc(5 * sizeof(int));</code>
10	Memory leak - ptr5 malloc'd twice	80	Runtime Error	<code>free(ptr5);</code> before second malloc
10	Function average() called with 2 arguments instead of 3	86	Semantic Error	<code>avg = average(70, 80, 90);</code>
10	Wrong age logic - > 18 should be eligible	94	Logical Error	<code>if(age >= 18)</code> <code>printf("Eligible\n");</code> <code>else</code> <code>printf("Not Eligible\n");</code> <code>;</code>
10	Wrong prime logic - isPrime set to 1 when divisible	104	Logical Error	<code>isPrime = 0;</code> <code>break;</code>
10	Undefined function displayReport()	116	Semantic Error	<code>Define displayReport()</code>
10	Undefined function calculateSalary()	118	Semantic Error	<code>Define calculateSalary()</code>
10	File not found - fopen returns NULL	122	Runtime Error	<code>if(fp != NULL)</code> <code>{ fscanf(...)</code> <code>};</code> <code>fclose(fp);</code> <code>}</code>
10	Missing colon after case 1	129	Syntax Error	<code>case 1:</code>
10	Missing break in case 2	137	Logical Error	<code>Add break</code> <code>after case 2</code>
10	Multi-character constant 'AB'	142	Type Error	<code>char grade = 'A';</code>
10	Array index out of bounds - scores[3] for size 3	149	Runtime Error	<code>int scores[4];</code>

Question No.	Error	Line No.	Error Type	Error Correction
10	Array index out of bounds - i<=3 for size 3	153	Runtime Error	<code>for(int i=0; i<3; i++)</code>
10	Wrong area formula - uses 2 (circumference)	162	Logical Error	<code>area = 3.14 * radius * radius;</code>
10	Semicolon after while makes empty loop	170	Logical Error	<code>while(choice > 0) { choice--; }</code>
10	continue outside loop	174	Syntax Error	Remove or place inside loop